

QUESTION 1

(a) Describe the main features of amylose visible in Fig. 1.1 [2]

- Ref to α (1, 4) glycosidic bonds between glucose units
- Ref to helical structure, with 6 glucose units per turn

(b) Explain how the structure of amylose makes it suitable for its function. [2]

- Ref to large amylose, thus insoluble in water, does not exert osmotic influence in the cell
- Ref to hundreds to thousands of glucose, thus a large source of carbon
- Ref to α (1, 4) glycosidic bonds, being easily hydrolysed
- Ref to amylose being helical and thus compact
- Ref to anomeric carbon involved in glycosidic bond formation, thus unreactive / chemically-stable

(c) Suggest how double helix structure of amylose is formed. [1]

Ref to via hydrogen bonds between adjacent amylose molecules

(d) (i) State two ways in which the structure of collagen differs from the structure of starch. [2]

	Collagen		Starch
1a	Made up of amino acids	1b	Made up of α -glucose
2a	Linked by peptide bonds	2b	Linked by α (1,4) glycosidic bonds
3a	Triple helix tropocollagen	3b	Helical amylose, highly branched amylopectin
4a	ref to unbranched	4b	Ref to branched

(ii) Explain how one of the differences stated in (d)(i) allows collagen to perform its function. [2]

- Ref to glycine-X-Y, glycine with small R group can fit in / packed tightly together
- Ref to providing high tensile strength
- Ref to extensive hydrogen bonds within tropocollagen, between NH group of Gly in one α -chain and CO group of another amino acid in next α -chain
- Ref to providing high tensile strength

[Total: 9]

QUESTION 2

With reference to Fig. 2.1,

(a) (i) explain why calcium ions do not pass through the phospholipid bilayer. [2]

- Calcium ions are water soluble/ hydrophilic
- unable to pass through hydrophobic core of the phospholipid bilayer

(ii) state and describe the process by which calcium ions are moved across the membrane. [4]

- Active transport
- Intracellular Ca^{2+} binds to the binding sites of the calcium pump/ carrier protein
- Upon binding of Ca^{2+} , carrier protein undergoes conformational change
- Ca^{2+} is transported against the concentration gradient from cytoplasm to outside the cell
- Ref. to usage of ATP as an energy source

(b) With respect to the fluid mosaic model, discuss how high temperature affects membrane permeability. [3]

- At high temperature, kinetic energy of the hydrocarbon chains of the phospholipids increases, and increase lateral movements
- Ref. to overcoming hydrophobic interactions between phospholipids, increase transient gaps between adjacent phospholipids
- Ref. to correct explanation following denaturation of membrane proteins at high temperature
- Ref. to increase membrane permeability

[Total: 9]

QUESTION 3

(a) State the Central Dogma of Molecular Biology. [3]

- unidirectional flow from DNA to RNA to polypeptide
- DNA transcribed to form RNA
- mRNA translated to form polypeptides

(b) Explain how a molecule of telomerase synthesises additional lengths of DNA. [4]

- template RNA binds to region X
- in the active site of telomerase
- free nucleotides bind to DNA template via hydrogen bonds
- complementary base pairs such as to A–T and C–G and U–A
- phosphodiester bonds formed between nucleotides
- telomerase moves in the direction of the arrow

(c) Explain why the action of telomerase challenges the Central Dogma of Molecular Biology. [2]

- RNA acts as a template
- reverse flow of genetic information

(d) Suggest why prokaryotes do not have telomerase. [2]

- prokaryotes have, circular DNA
- prokaryotes do not have telomeres

[Total:11]



QUESTION 4

(a) Compare the structures of SARS-CoV-2 and influenza virus.

[3]

- Both contain RNA
- Both have viral envelope with glycoproteins on its surface
- There are 8 segments of RNA in influenza but 1 RNA for SARS-CoV-2
- HA glycoprotein is embedded on viral envelope of SARS-CoV-2 whereas S glycoprotein is embedded in the viral envelope of COVID-19

(b) Describe how the SARS-CoV-2 virus enters a host cell.

[3]

- S glycoprotein on the viral membrane binds to ACE2 and PRS2 receptors on the host cell membrane
- The virus enters the host cell by receptor-mediated endocytosis
- Ref. to subsequent viral envelope fusing with vesicle membrane and releasing the nucleocapsid into cytoplasm

(c) There are at least three variants of the SARS-CoV-2 virus since its discovery in 2019.

Identify and describe one type of variation in viral genomes that may lead to the formation of new strains of the SARS-CoV-2 virus.

[3]

- Ref. to antigenic shift / antigenic drift
- (for antigenic shift) Reassortment of viral genome with that of a different antigenic type results in the formation of a new COVID-19 strain
- (for antigenic drift) Gradual accumulation of minor mutations results in changes to the genes for the glycoprotein receptors
- Ref. to change of 3D conformation / shape of proteins
- Ref. to lack of proofreading mechanism associated with the enzyme of the virus

[Total: 9]



QUESTION 5

(a) Explain how this mutation causes the HbS to form fibres. [3]

- ref. to change in β -globin gene and amino acid
- ref. to HbS sticking together

(b) Suggest why PCR may be needed before the extracted DNA can be profiled. [2]

- ref. to small amount of DNA to be amplified

(c)(i) Describe the role of primers in PCR. [2]

- ref. to binding to DNA to be amplified

(ii) Explain how the use of two specific primers allows the detection of the normal, sickle cell anaemia and SCT genotypes. [3]

- ref. to binding of specific primers for the respective genotypes

[Total: 10]

QUESTION 6

(a) Outline how cancer is caused. [3]

- Ref. to accumulation of mutations
- Ref. to the dysregulation of cell cycle checkpoints
- Ref. to uncontrolled cell division

(b) Describe the normal role of the *BRCA2* gene. [3]

- Ref. to the tumour suppressor gene *BRCA2* coding for the BRCA2 protein

The tumour suppressor protein / BRCA2 protein

- takes part in the cell-signaling pathways to inhibit the cell cycle
- halts cell division if DNA is damaged

(c) Discuss the extent to which Fig. 6.1 provides evidence that a faulty *BRCA2* allele increases the risk of a person developing cancer. [4]

- Ref. to the idea that the overall data in Fig. 6.1 is inconclusive

Data that shows *BRCA2* allele might increase the risk of a person developing cancer

- Individual 3 or 4 may have had the, *BRCA2* allele / Any individual from 8 to 11 may have inherited, *BRCA* allele, from 3 or 4
- Individual 15 has cancer, *BRCA2* allele

Data that shows *BRCA2* allele might not increase the risk of a person developing cancer (i.e. data is inconclusive)

- Individual 8 or 11, has *BRCA2* allele, but does not have cancer
- No evidence / unknown, that individuals (apart from 15) with cancer, have *BRCA2* allele

[Total: 10]



QUESTION 7

(a) Explain the term *epistatic gene interaction* in this context.

[2]

- The expression of *tt* masks the expression of alleles at the *M/m* locus
- Ref to presence of *T* allele being required for expression of red / black hair /ORA

(b) A couple with black hair has three children, each with a different hair colour.

Draw a genetic diagram to explain these results.

[4]

Parental phenotype		Both heterozygous black hair parents				}	;
Parental genotype		MmTt		x	MmTt		
Gametes		MT	Mt	mT	mt		
		Female gametes				}	;
		MT	Mt	mT	mt		
Male Game tes	MT	MMTT	MMTt	MmTT	MmTt		
	Mt	MMTt	MMtt	MmTt	Mmtt		
	mT	MmTT	MmTt	mmTT	mmTt		
	mt	MmTt	Mmtt	mmTt	mmtt		
Offspring genotypic ratio		9 M ₋ T ₋ :	3 mmT ₋ :	3 M ₋ tt:	1 mmtt	}	;
Offspring phenotypic ratio		9 black hair:	3 red hair:	4 (3+1) albino		}	;



(c)(i) Complete Table 7.1 to show the expected numbers of individuals with each phenotype. [1]

hair colour	observed number of individuals	expected number of individuals
black	579	540
red	165	180
albino	216	240

(c)(ii) Using the formula above, calculate the χ^2 value for the observed results. Show your working clearly. [1]

$$\chi^2_{\text{calc}} = (540-579)^2/540 + (180-165)^2/180 + (240-216)^2/240 \\ = 6.47$$

(c)(iii) Explain the conclusion that may be drawn from your χ^2 value in (c)(ii). [3]

- At $df=2$, $\chi^2_{\text{calculated}} = 6.47 > \chi^2_{\text{critical}} = 5.99$;
- there is less than 5% probability that there is difference between observed and expected results is due to chance ;
- Thus the distribution of hair types shows significant difference from 9:3:4/ does not follow the predicted ratio of 9: 3: 4/ there is a larger distribution of dark haired individuals at the equator;

[Total: 11]

QUESTION 8

(a) Explain the roles of proteins involved in the light-dependent reaction of photosynthesis. [2]

- enzymatic role
- electron transport chain
- active transport for chemiosmosis
- photosystems with ability to carry pigment for absorption of light

(b)(i) Describe and explain the pattern of results obtained between 0.25 mol dm^{-3} and 1.25 mol dm^{-3} of sodium hydrogencarbonate solution. [4]

- increasing trend because CO_2 concentration is limiting
- subsequent plateau because CO_2 concentration is no longer limiting

(ii) Suggest reasons for this negative value. [2]

- low photosynthesis as low CO_2 , respiration uses up oxygen

(iii) To minimise temperature changes, the student decided to use an LED lamp as a light source. LED lamps release very little heat energy.

Explain the importance of minimising temperature changes in this experiment. [3]

- so that CO_2 is the only independent variable
- high temperature may denature enzyme

[Total: 11]



QUESTION 9

(a) (i) State why the two bee species share the first name *Bombus*. [1]

Same genus

(a) (ii) Describe how it is possible to confirm, over a longer period of time, whether *Bombus pratorum* and *Bombus terrestris* belong to different species. [2]

- Check if *Bombus pratorum* and *Bombus terrestris* interbreed
- Determine if offspring are viable and fertile
- Conclusion, i.e. if offspring are infertile, then they are different species

(b)(i) Name the process by which new species are formed within the same geographical location. [1]

- Sympatric speciation

(b)(ii) Using Fig. 9.1 and Table 9.1, and the information given, suggest how an ancestral species might have evolved into the two species, *B. pratorum* and *B. terrestris*. [5]

- Ref to seasonal / temporal difference in breeding of males and queens in different populations with data quoted from Fig. 9.1
- Ref to behavioural difference / visit different types of flowers / feed at different times / feed on different food types with data quoted from Table 9.1
- Ecological isolation
- Gene flow restricted
- Natural selection, resulting from different adaptations under different selection pressures
- Accumulation of sufficient reproductive isolating mechanisms, genetic diversity and adaptations
- Reproductive isolation

[Total: 9]

QUESTION 10

(a) Describe the innate immune response. [2]

- combats microbes immediately upon infection / mediates the initial protection against infections
- ref. to specific named cells and molecules involved, e.g. macrophages, chemokines, cytokines, complement system

(b) Explain which immune responses are impaired in Patients A and B [3]

- Patient A - innate / innate and adaptive, Patient B - adaptive
- ref. to non functional innate immune response, leading to high number / exponential increase of microorganism count
- ref. to functional innate but non functional adaptive, hence slow rise at the start, microorganism count continues to rise

[Total: 5]



QUESTION 11

(a) With reference to the information provided, suggest and explain how climate change scientists can estimate atmospheric carbon dioxide concentrations 10 000 years ago. [2]

- ref. to how carbon dioxide air bubbles is a good sample / indicator of atmospheric carbon dioxide levels
- ref to extracting ice samples from different depths of ice sheets and determine the age of the ice samples

(b) Use Fig. 11.2 to comment on changes in temperature over the last 800 000 years. [2]

- Rise and fall of ratio of ^1H to ^2H corresponds with the percentage concentration of carbon dioxide
- Carbon dioxide is a greenhouse gas, increased concentration in the atmosphere lead to more heat being trapped, rise and fall of temperature as the concentration of carbon dioxide changes

(c) Explain why the data in Fig. 11.2 do not provide enough evidence to conclude that increased concentrations of greenhouse gases cause climate change. [2]

- Carbon dioxide is not the only greenhouse gas / unable to identify which gas is causing the change
- ref. to temperature not the only factor in climate change
- ref. to correlation does not imply causation

[Total: 6]

